



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

OXIDIZER NO.2 (OX-2202)

Docu. No. : 14780-8120-PS-OX2202
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 10

TITLE :

OXIDIZER NO.2

(OX-2202)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-OX2202**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	MAY 22, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A1	OCT 2, 2013	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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VESSELS

PROJECT SPECIFICATION

963765 — 301

SHEET 3 / 10

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	27APR12	TS	HDT				

General Notes

- A. Vessels shall be constructed in accordance with the following unless otherwise specified on the individual vessel specification sheet.

Standards:

UOP Standard Specification 3-11

- B. Although the resulting inside diameter may be slightly different than specified, vessels without trays or packing, may be fabricated from pipe.
- C. The dimensioned location of some of the vessel level nozzles may be noted as min (minimum) or max (maximum). It is the responsibility of the contractor to adjust these dimensions to suit the specific level requirements specified on the UOP Piping and Instrument Diagram(s) and on UOP Standard Drawing 8-121. When additional nozzles are required for level instrumentation, the contractor shall specify them consistent with the size, class and facing requirements specified in the UOP Project Specification 301-Vessels and on the UOP Piping and Instrument Diagram(s).
- D. When system of measurement is metric, all dimensions are in millimeters unless noted otherwise.
- E. Contractor shall not locate TI thermowells in a downcomer. Contractor shall also ensure that TI thermowells are located to avoid interference with downcomers, weirs, distributors, collectors, baffles, chimneys, traps and all other vessel internals.
- F. Suppliers (or equal):

Spray Nozzle:

BETE Fog Nozzle Inc.
www.bete.com

Profile Wire Screens:

Johnson Screens
www.johnsonscreens.com

Nagaoka International Corporation
www.nagaokajapan.co.jp

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-11, 3-15, 3-35, 9-11

UOP Standard Drawing: 3-115, 3-116, 3-122, 3-123, 3-144, 3-183, 3-190, 3-264, 3-266, 3-313, 3-322, 8-121

UOP Project Specification: 963765-106, 963765-307



VESSELS

SERVICE OXIDIZER NO. 2 ITEM NO OX-2202

Design	INT	0.45 kg/cm2 (g) @ 130°C
Conditions	EXT	0.01 kg/cm2 (g) @ 82°C
At Top		Metal Temperature (Min) 15°C
Operating Conditions		0.187 kg/cm2 (g) @ 82°C

Radiograph	MR
Postweld Heat Treat	MR
Joint Efficiency	MR %

Material Specifications	
Heads	A240 TYPE 304/ 304L (DUAL STAMP)
Shell	A240 TYPE 304/ 304L (DUAL STAMP)
Internals	A240 TYPE 304/ 304L (DUAL STAMP) (EXCEPT AS NOTED)

Shell	Thickness Corrosion Req'd By + Allowance Code-mm mm (Min)
	1.5

Heads	1.5
-------	-----

Vessel Heads

Top Head:	MR
Bottom Head:	SLOPE UP
Accessories Applied by Fabricator	
Ladder & Platform Clips	YES
Insulation Clips & Rings (NOTE 2) F(80)	YES
Vessel Support (NOTE 3)	NO

Nozzles and Manways

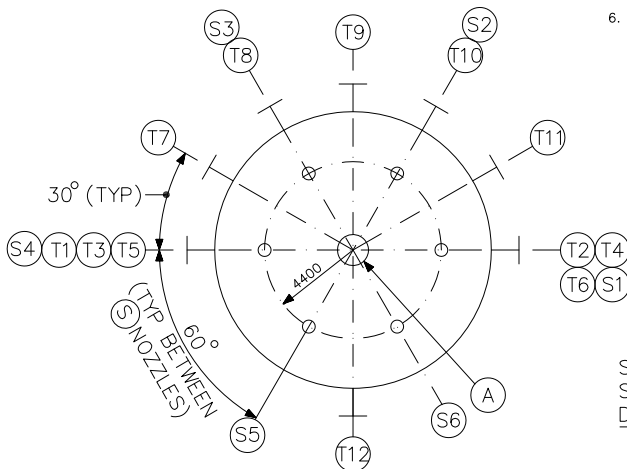
Mark	No	Size Inches	Service
A	1	30	VAPOR OUTLET
B	1	24	CIRCULATION OUTLET
D	2	18	AIR INLET
F	1	16	FEED/CIRCULATION INLET
G	2	3	LIC
H	1	14	PUMPOUT
K	1	3	NITROGEN
L	2	3	LRC
M	MR	36	MANWAY
N	NOTE 11		PRESSURE RELIEF EQUALLY SPACED ON ROOF
P	1	2	PI
Q	2	3	LEVEL
R	NOTE 4		VACUUM RELIEF EQUALLY SPACED ON ROOF
S1-S6	6	3	CONDENSATE SPRAY
T1-T12	12	2	TAH
U	1	6	TI (STILLING WELL)
V	1	24 ID	MANWAY (OBSERVATION)
W	1	3	PI
X	1	3	FOR PUMPOUT

Class - ALL EXTERNAL CONNECTIONS SHALL BE ASME CL150 EXCEPT FOR NOZZLES (C), (L), AND (Q) TO BE ASME CL300

Facing - RAISED FACE

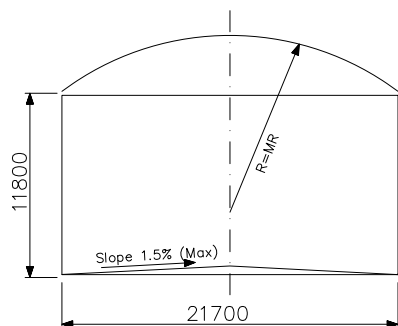
Normal Liquid Level = 10300 ABOVE BOTTOM
NOTE 7 (AERATED)

Specific Gravity = 0.81 (AERATED)



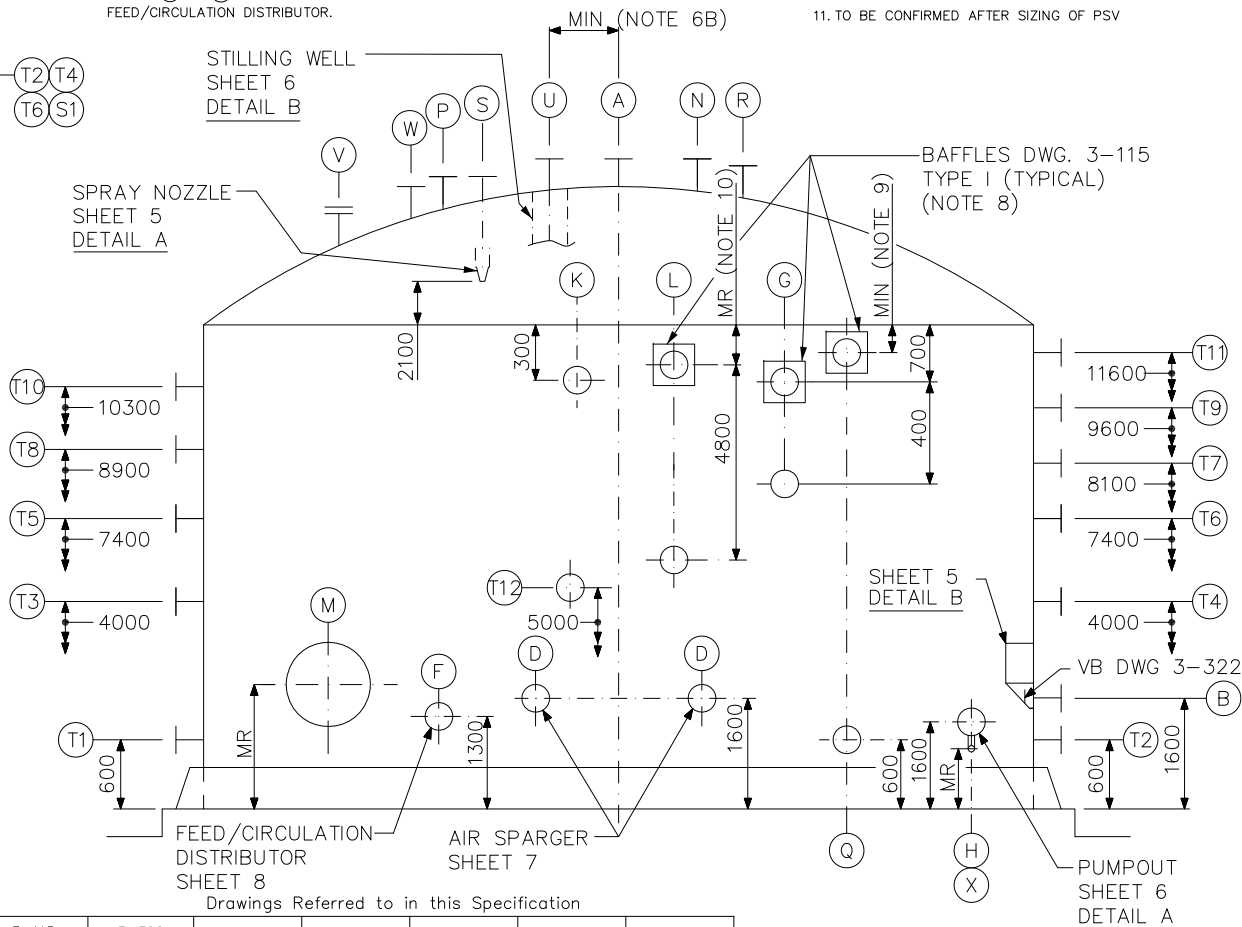
NOZZLE ORIENTATIONS

ORIENTATIONS OF NOZZLES NOT SHOWN SHALL BE MR EXCEPT AS NOTED IN NOTE 6.



NOTES:

- VESSEL SHALL BE DESIGNED AND CONSTRUCTED IN ACCORDANCE WITH API 620. THE SPECIFIED DESIGN TEMPERATURE EXCEEDS API RECOMMENDED TEMPERATURE IN ORDER TO BE CONSISTENT WITH THE MAXIMUM EMERGENCY TEMPERATURE THAT MAY BE ATTAINED BY THE PROCESS FLUIDS. DUE TO THE NATURE OF THE MATERIAL IN THE TANK, STAGNANT AREAS MUST BE AVOIDED WHEN APPLYING API 620 RULES.
- INSULATION AND WEATHERSHIELDING SHALL BE IN ACCORDANCE WITH UOP STANDARD SPECIFICATIONS 9-11. THE TYPE OF INSULATION MATERIAL, THICKNESS, WEATHERPROOFING, AND ATTACHMENT METHODS SHALL BE DETERMINED BY THE OWNER/CONTRACTOR. (FIRE INS: SECT. 2.7) REFER TO PROJECT SPECIFICATION 963765-907.
- THE OXIDIZER BOTTOM SHOULD NOT COME IN CONTACT WITH ANY CHLORIDE. A POSITIVE BARRIER MUST BE PROVIDED BETWEEN TANK FLOOR AND SUPPORT MATERIAL.
- VENDOR TO CONFIRM THAT NOZZLE (R) IS COMPATIBLE WITH VACUUM RELIEF DEVICE SELECTED. VACUUM RELIEF DEVICE SHALL BE PROVIDED BY MECHANICAL TEAM.
- ALL INTERNALS SHALL BE DESIGNED TO FIT THROUGH NOZZLE (M).
- CUSTOMER/CONTRACTOR TO ORIENT ALL NOZZLES EXCEPT AS SHOWN IN NOZZLE ORIENTATION VIEW. NOZZLES SHALL BE LOCATED AS PER NOTES BELOW:
 - ORIENT NOZZLE (B) 90° FROM NOZZLE (D).
 - LOCATE (U) SUCH THAT STILLING WELL DOES NOT INTERFERE WITH AIR SPARGER.
 - LOCATE (H) BETWEEN AIR SPARGER BRANCHES TO PREVENT INTERFERENCE BETWEEN AIR SPARGER AND PUMPOUT PIPING (SEE SHEET 6).
 - LOCATE (H) AWAY FROM (B).
 - ORIENT (F) AND (D) SUCH THAT THE AIR SPARGER DOES NOT INTERFERE WITH THE FEED/CIRCULATION DISTRIBUTOR.



Drawings Referred to in this Specification

PROJECT SPECIFICATION

963765 - 301 SHEET 4 / 10

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	20APR12	STS	RJS				
1	03JUL12	JG	HDT				
2	07SEP12	JG	HDT				

- ALTERNATE OPERATION CONDITION EXISTS WITH LIQUID LEVEL EQUAL TO TOP VESSEL TANGENT LINE. LIQUID SPECIFIC GRAVITY FOR ALTERNATE OPERATION CASE = 0.87.
- IF REQUIRED, THE BAFFLE LENGTH MAY BE TRIMMED FROM DIMENSIONS SHOWN IN DWG. 3-115 TO ALLOW BAFFLE TO FIT WITH TOP VESSEL HEAD.
- PREFERRED DISTANCE TO THE TANGENT LINE IS 50 mm. THIS DISTANCE SHALL BE MINIMIZED FOR OPERATING CONDITION WHEN FILLING LIQUID TO TOP TANGENT LINE. MAXIMUM DISTANCE SHALL BE 240 mm.
- PREFERRED AND MINIMUM DISTANCE TO THE TANGENT LINE IS 200 mm. MAXIMUM DISTANCE SHALL BE 240 mm.
- TO BE CONFIRMED AFTER SIZING OF PSV

3-115

3-322



VESSELS

VESSEL DETAILS

SERVICE	OXIDIZER	ITEM NO	OX-2201 / OX-2202
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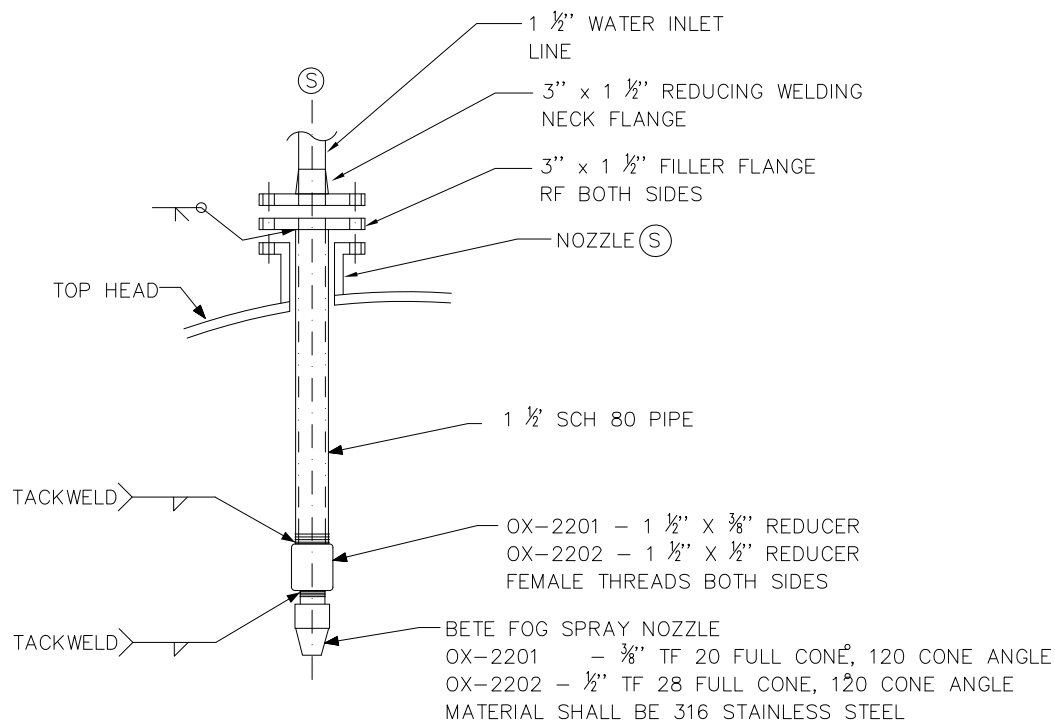
NOTES:

1. MINIMUM REQUIRED SUPPLY PRESSURE AT WATER SPRAY NOZZLE IS $3.6 \text{ kg/cm}^2(\text{g})$.

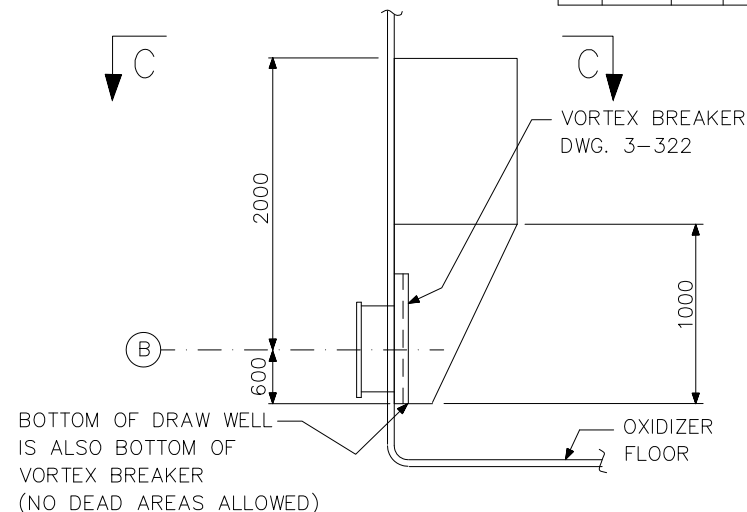
PROJECT SPECIFICATION

963765 - 301 SHEET 5 / 10

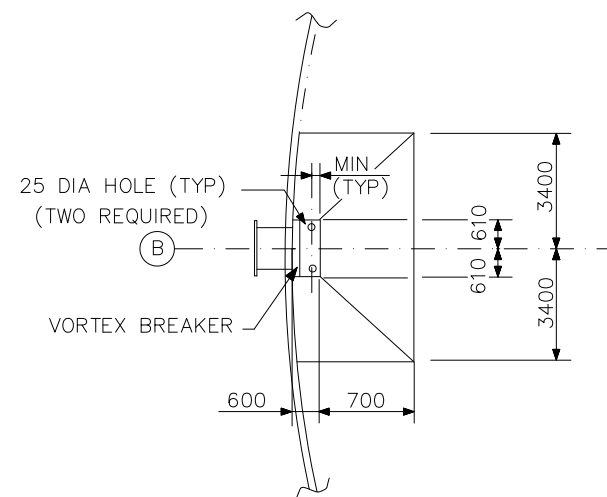
REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	20APR12	STS	RJS				



DETAIL A
WATER SPRAY NOZZLE
6 - REQUIRED
(NOTE 1)



DETAIL B



SECTION C-C

Drawings Referred to in this Specification

3-322

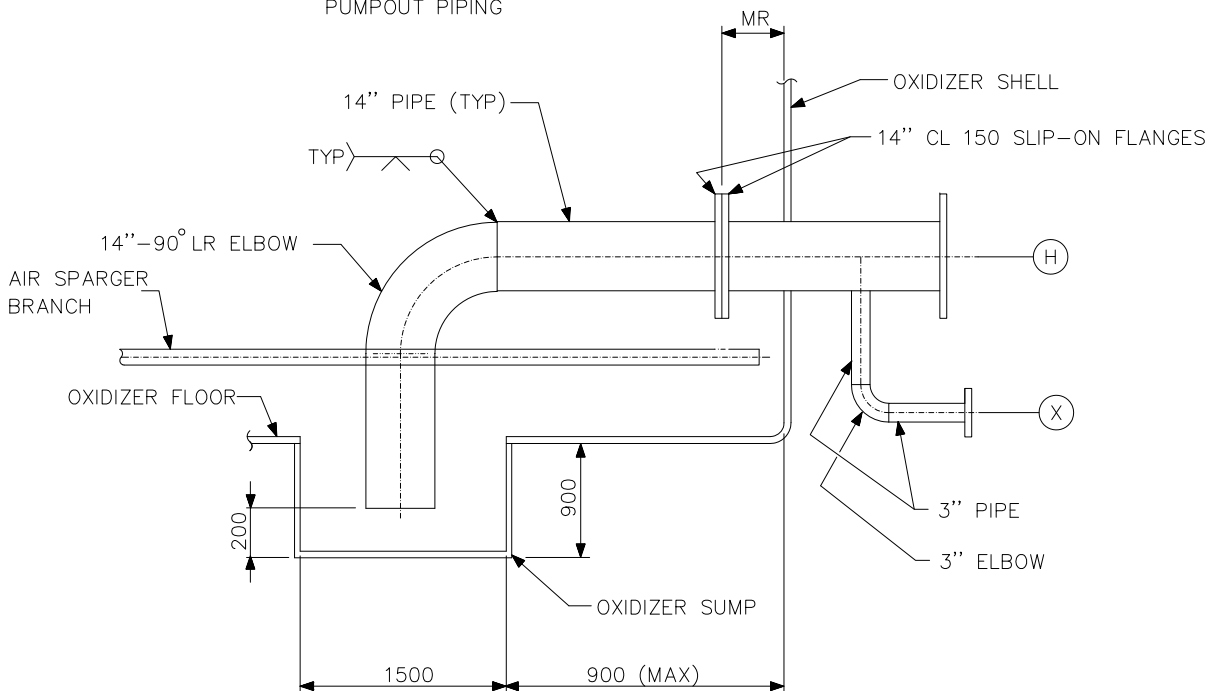


VESSEL DETAILS

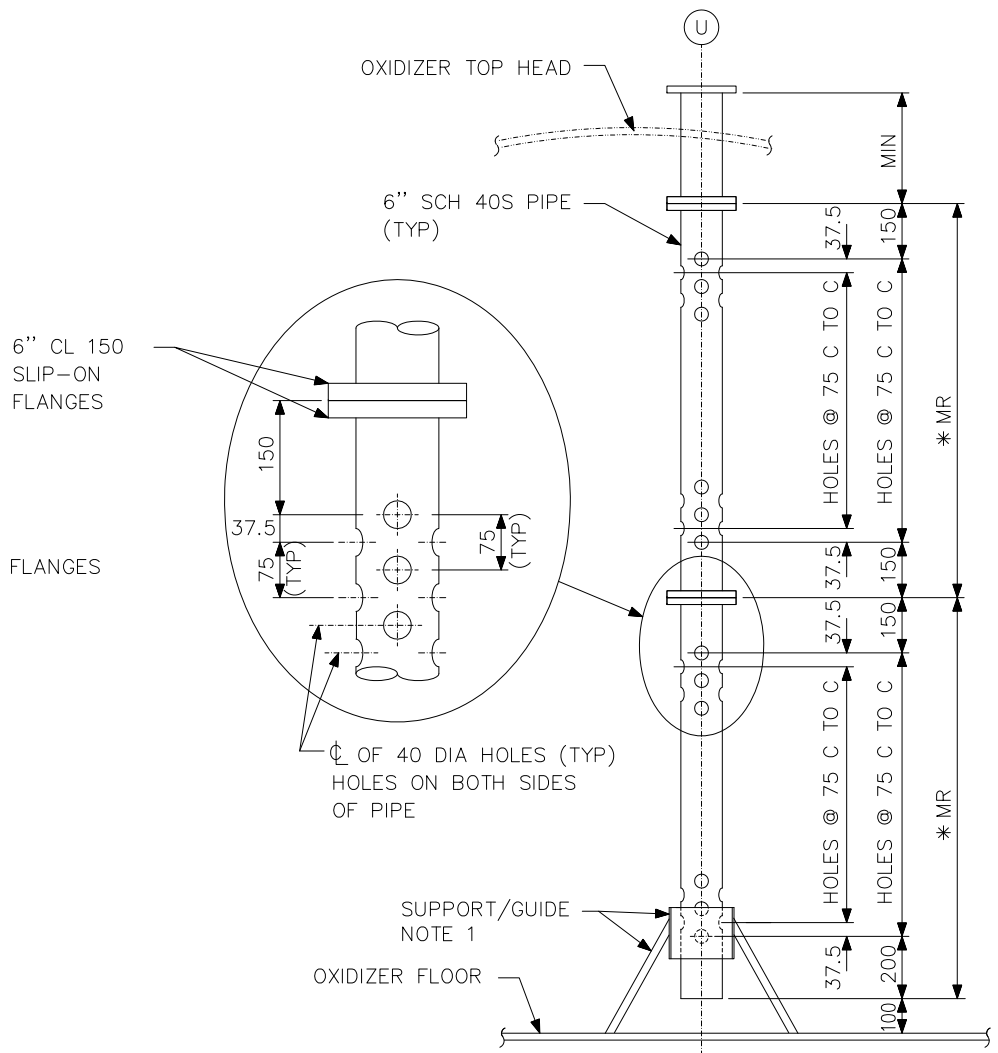
SERVICE	OXIDIZER	ITEM NO	OX-2201 / OX-2202
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1. PIPE STILLING WELL SUPPORT/GUIDE BY CONTRACTOR.
THE SUPPORT SHALL BE DESIGNED TO ALLOW FOR SOME
DIFFERENTIAL EXPANSION BETWEEN THE TANK AND INTERNALS.

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	20APR12	STS	RJS				



DETAIL A



DETAIL B

PIPE STILLING WELL

* PROVIDE TWO SECTIONS OF EQUAL LENGTH

Drawings Referred to in this Specification

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VESSELS

AIR SPARGER

SERVICE OXIDIZER ITEM NO 0X-2201 / 0X-2202

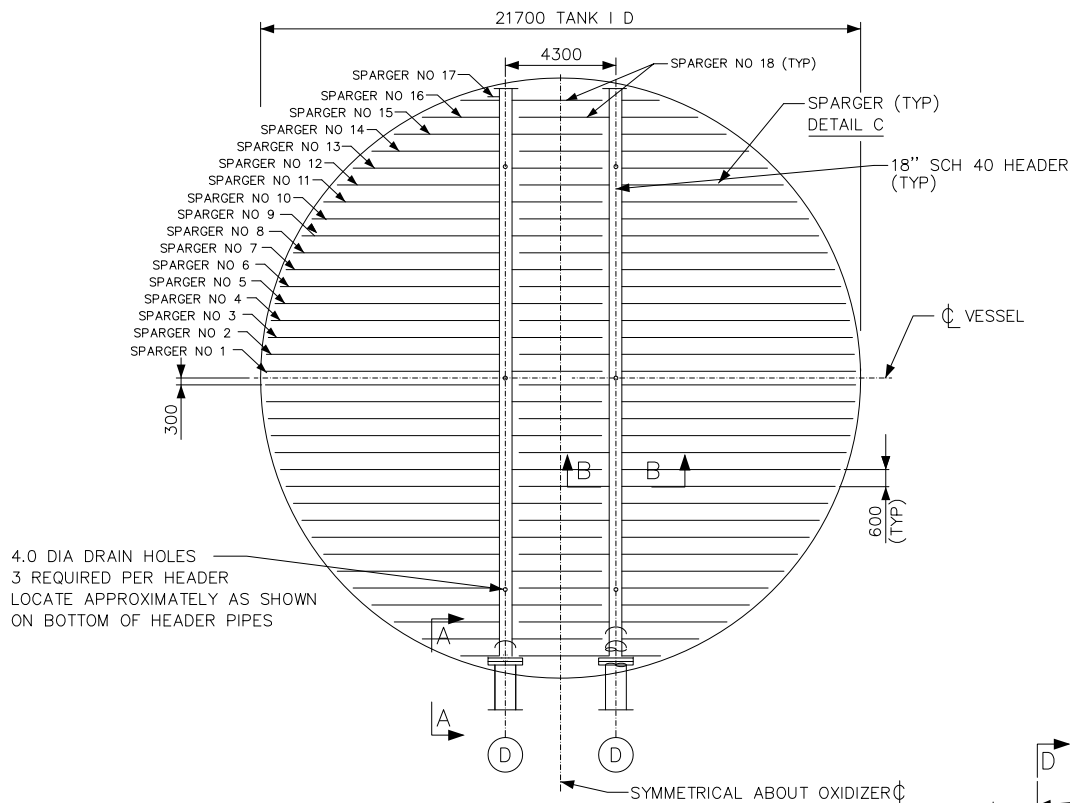
NOTES:

- 1.SPARGER SUPPORT BY CONTRACTOR. SPARGERS MAY BE SUPPORTED OFF OF HEADER PIPE OR OXIDIZER FLOOR. SUPPORT MATERIAL SHALL MATCH MATERIAL FROM WHICH IT IS SUPPORTED.
- 2.PROVIDE ONE DRAIN HOLE ON BOTTOM OF EACH SPARGER. DRAIN HOLE DIAMETER TO MATCH SPARGER HOLE DIAMETER SHOWN IN DETAIL C.

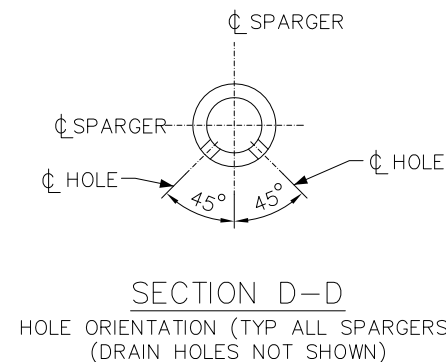
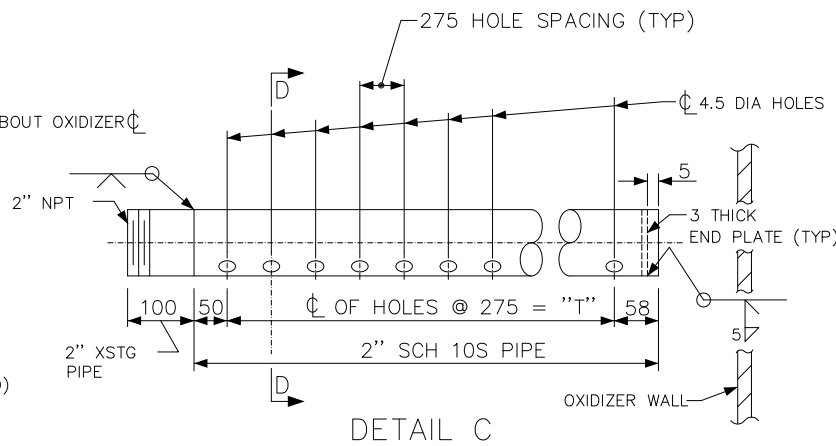
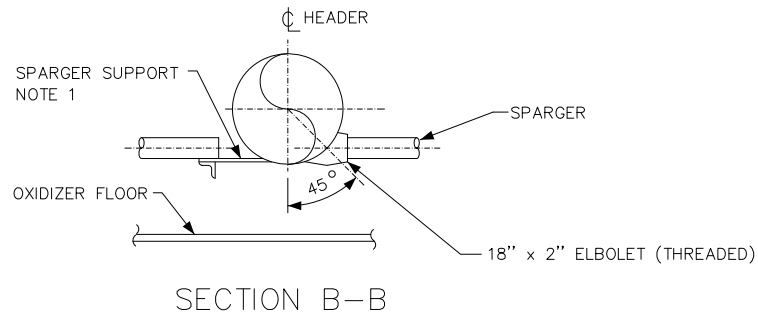
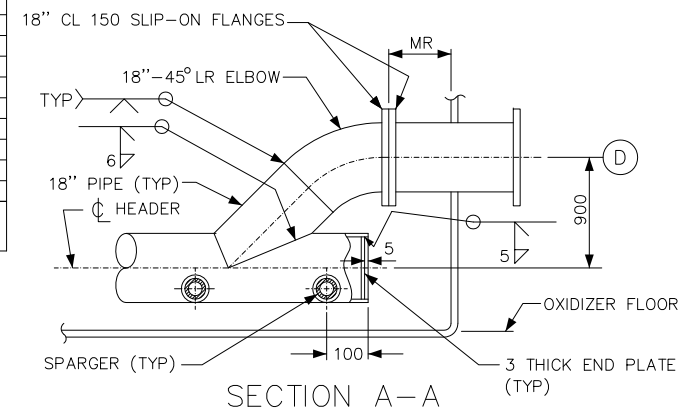
PROJECT SPECIFICATION

963765 - 301 SHEET 7 / 10

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	20APR12	STS	RJS				



SPARGER NUMBER	NUMBER REQUIRED	NUMBER OF HOLES ON EACH SPARGER	"T"
1	4	60	7975
2	4	60	7975
3	4	60	7975
4	4	58	7700
5	4	58	7700
6	4	56	7425
7	4	54	7150
8	4	52	6875
9	4	50	6600
10	4	48	6325
11	4	46	6050
12	4	42	5500
13	4	38	4950
14	4	34	4400
15	4	28	3575
16	4	20	2475
17	4	12	1375
18	34	26	3300
TOTAL NUMBER OF HOLES = 3988 (NOT INCLUDING DRAIN HOLES)			



SECTION D-D
HOLE ORIENTATION (TYP ALL SPARGERS)
(DRAIN HOLES NOT SHOWN)



VESSELS

FEED / CIRCULATION DISTRIBUTOR

SERVICE OXIDIZER ITEM NO OX-2201 / OX-2202

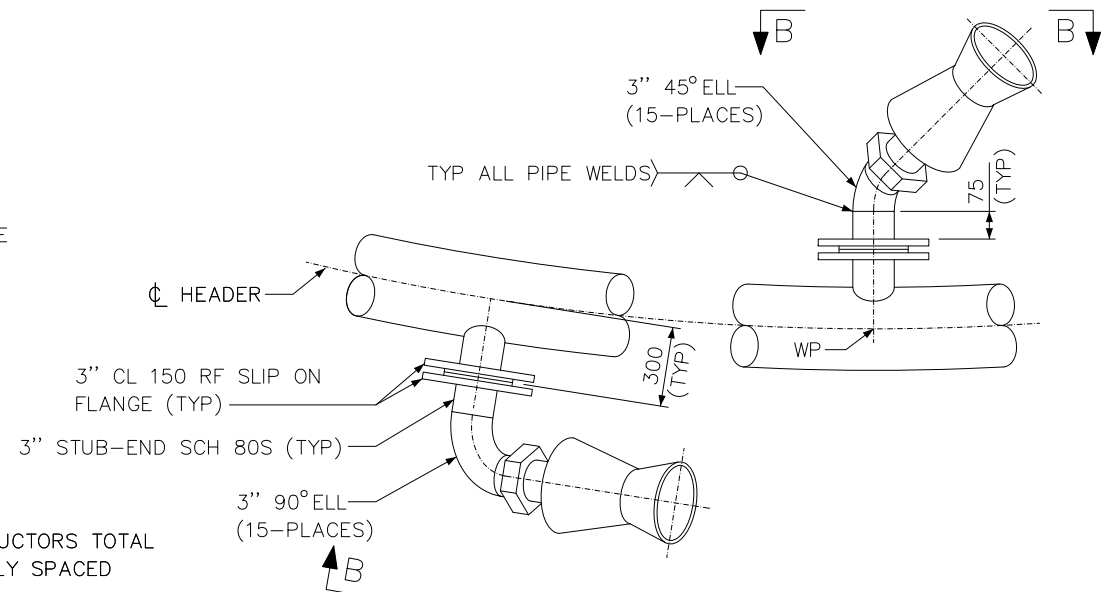
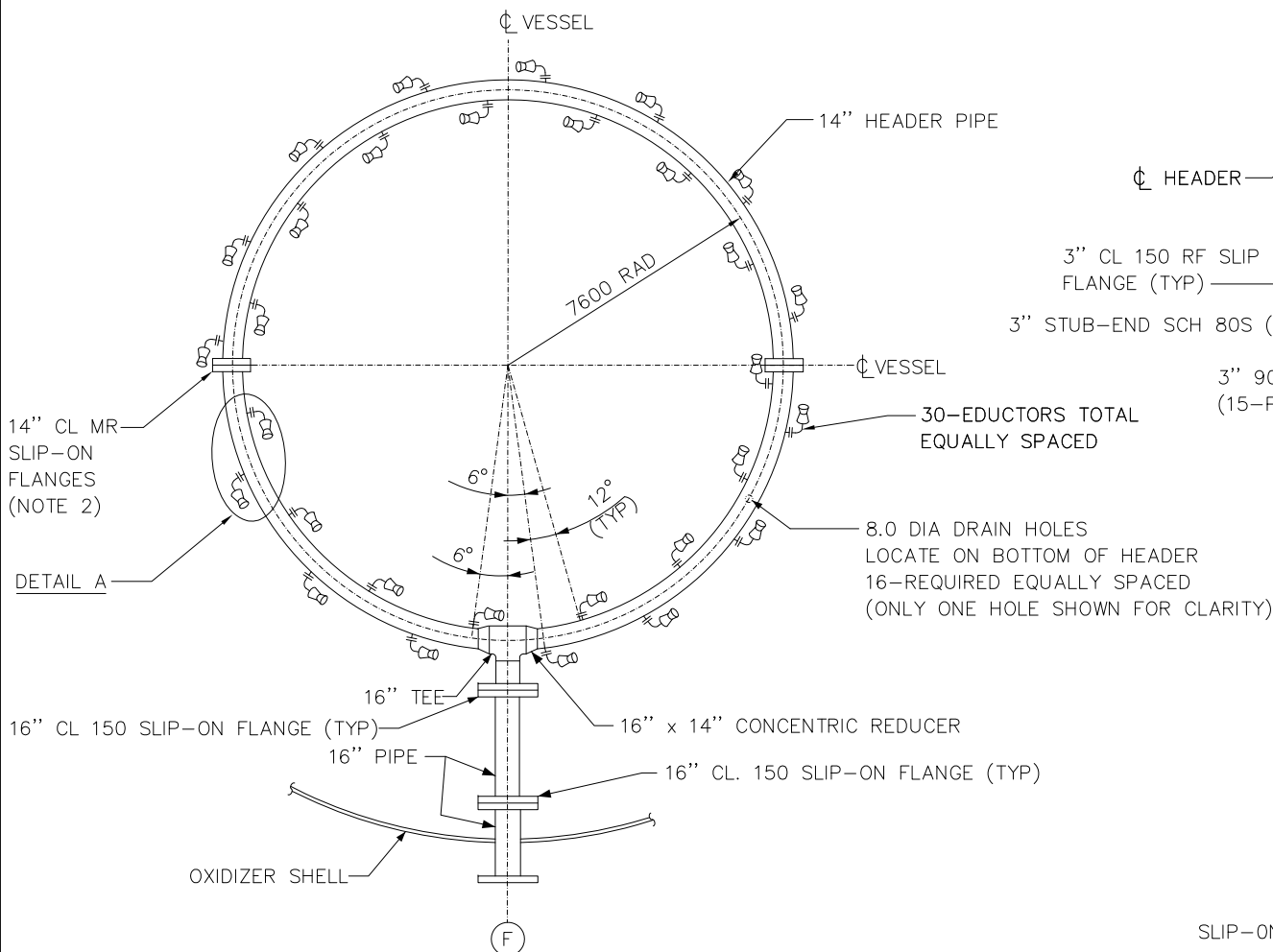
NOTES:

1. DISTRIBUTOR SUPPORT BY CONTRACTOR.
2. NUMBER AND LOCATION OF FLANGES SHALL BE MR (AS REQUIRED TO FIT DISTRIBUTOR THROUGH MANWAY AND FOR EASE OF INSTALLATION). FLANGES SHALL NOT INTERFERE WITH EDUCTORS OR EDUCTOR SPRAY.
3. ALL PIPE SCHEDULES TO BE MR EXCEPT AS NOTED.

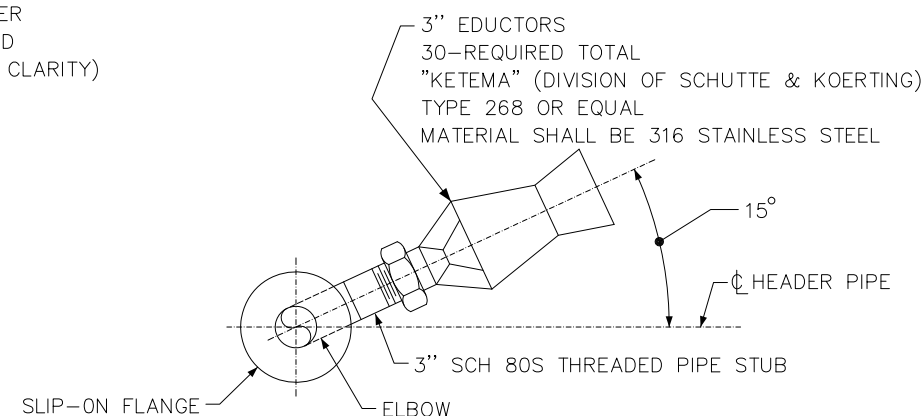
PROJECT SPECIFICATION

963765 - 301 SHEET 8 / 10

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	20APR12	STS	RJS				



DETAIL A B



VIEW B-B

FINAL POSITIONING OF EDUCTORS SHALL BE DONE DURING INSTALLATION AT SITE (TYP FOR ALL EDUCTORS)

Drawings Referred to in this Specification

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Revision
Indication



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VACUUM VENT

PROJECT SPECIFICATION

963765 — 989

SHEET 9 / 10

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	06JUN12	JLS	HDT				

UOP Phenol Process Unit Oxidation Section

Provide vacuum vents (Protectoseal Series 6240 V or equivalent) in accordance with the following. Vacuum vents shall have weighted pallets for vacuum relief.

Item Number	PVRV-1401A/B
Service	OX-2202, Oxidizer No. 2
Number	Two
Valve Size:	8 Inches
Connection:	ASME CL 150 RF Flange
Materials	
Body:	316 Stainless Steel
Valve Trim:	316 Stainless Steel
Tank	
Diameter, meters	21.7
Height, meters	11.8
Normal Pump In Rate, m ³ /hr	275
Normal Pump Out Rate, m ³ /hr	275
Maximum Pump Out Rate, m ³ /hr	1419
Normal Operating Pressure, kg/cm ² (g)	0.211
Tank Internal Design Pressure, kg/cm ² (g)	0.45
Vacuum Relief Setting, kg/cm ² (g)	0.0088 (Below Atmospheric Pressure)
Tank External Design Pressure, kg/cm ² (g)	0.01
Vacuum Relieving Rate	Per API-2000
Fluid in contact with valve:	Air / Cumene mixture
Flash point, °C	44

Other specifications
referenced in this
specification:

UOP Standard Specification:

UOP Standard Drawing:

UOP Project Specification:

Revision
Indication



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VACUUM VENT

PROJECT SPECIFICATION

963765 — 989

SHEET 10 / 10

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	06JUN12	JLS	HDT				

Notes:

1. Vacuum vents will be mounted on top of tank.
2. Oxidizer tanks are designed per API-620, *Design and Construction of Large, Welded, Low-pressure Storage Tanks*
3. Vacuum relieving rates shall be per API-2000, *Venting Atmospheric and Low-pressure Storage Tanks*
4. Vacuum vent vendor shall confirm that valve size and number of valves are sufficient for flow rates and pressures indicated.
5. Vacuum vent vendor shall ensure that weighted pallet is designed for tank internal design pressure and that pallet will not be distorted with pressure at tank internal design pressure.
6. Vacuum vents shall be provided with all accessories required for proper operation in an outdoor location.
7. Normal pump rates shown are net in or out of tank. A larger flow rate is circulated from tank immediately back into the tank. Vacuum relieving rate shall be based on the maximum pump out rate listed.
8. All vacuum vents shall normally be in operation. No spare vents are provided.

Supplier (or equal):

Protectoseal
www.protectoseal.com

Other specifications
referenced in this
specification:

UOP Standard Specification:

UOP Standard Drawing:

UOP Project Specification: